

Final Project - Product Price Comparison Application using Python



Estimated Time Needed: 1.5 hours

- In this final project you will be deploying multiple microservices to create an integrated application. These components consist of two backend microservices, one developed in Python and the other in Node.js, complemented by a front-end microservice.

Objectives:

After completing this lab, you will be able to:

- Deploy Python and Node.js backend microservices on IBM Cloud Code Engine.
- Deploy the frontend microservices on IBM Cloud Code Engine.

Part A: Deploying the Backend Microservices

1. Open the Code Engine CLI.
2. Deploy the microservice for Product Details, which provides API endpoints to retrieve product information.

build-source - https://github.com/ibm-developer-skills-network/dealer_evaluation_backend.git

build-context-dir - products_list

port - 5000

```
ibmcloud ce application create --name prodlist --image us.icr.io/${SN_ICR_NAMESPACE}/prodlist --registry-secret icr-secret --port 5000 --build-context-dir products_list --build-source https://github.com/ibm-developer-skills-network/dealer_evaluation_backend.git
```

Copy the deployment URL and save it in a notepad or other text editors.

Take a screenshot of the successful deployment and save it as product_details_deploy.png.

```
theia@theiadocker-lavanyas:/home/project$ ibmcloud ce application create --name prodlist
--image us.icr.io/${SN_ICR_NAMESPACE}/prodlist --registry-secret icr-secret --port 5000
--build-context-dir products_list --build-source https://github.com/ibm-developer-skill
s-network/dealer_evaluation_backend.git
Creating application 'prodlist'...
Submitting build run 'prodlist-run-230110-074851157'...
Creating image 'us.icr.io/sn-labs-lavanyas/prodlist:230110-1248-j11hw'...
Waiting for build run to complete...
Build run status: 'Running'
Build run completed successfully.
Run 'ibmcloud ce buildrun get -i prodlist-run-230110-074851157' to check the build run s
tatus.
Waiting for application 'prodlist' to become ready.
Configuration 'prodlist' is waiting for a Revision to become ready.
Ingress has not yet been reconciled.
Waiting for load balancer to be ready.
Run 'ibmcloud ce application get -n prodlist' to check the application status.
OK
https://prodlist.xj562h09ws5.us-south.codeengine.appdomain.cloud
```

Please note that if you encounter the error FAILED Wait failed for application 'prodlist', you can rename the application to prodlist1 and re-execute the command.

3. Deploy the microservice for Dealer Pricing, which provides API endpoints to retrieve dealer pricing information.

Note: Please use the below parameters for the deploy command

build-source - https://github.com/ibm-developer-skills-network/dealer_evaluation_backend.git

build-context-dir - dealer_details

port - 8080

name - dealerdetails

image - us.icr.io/\${SN_ICR_NAMESPACE}/dealerdetails

Copy the deployment URL and save it in a notepad or other text editors.

Take a screenshot of the successful deployment and save it as dealer_details_deploy.png.

```
theia@theiadocker-lavanyas:/home/project$ ibmcloud ce application create --name dealerdet
ails --image us.icr.io/${SN_ICR_NAMESPACE}/dealerdetails --registry-secret icr-secret --p
ort 8080 --build-context-dir dealer_details --build-source https://github.com/ibm-develo
per-skills-network/dealer_evaluation_backend.git
Creating application 'dealerdetails'...
Submitting build run 'dealerdetails-run-230110-075151354'...
Creating image 'us.icr.io/sn-labs-lavanyas/dealerdetails:230110-1251-h11hw'...
Waiting for build run to complete...
Build run status: 'Running'
Build run completed successfully.
Run 'ibmcloud ce buildrun get -i dealerdetails-run-230110-075151354' to check the build r
un status.
Waiting for application 'dealerdetails' to become ready.
Configuration 'dealerdetails' is waiting for a Revision to become ready.
Ingress has not yet been reconciled.
Waiting for load balancer to be ready.
Run 'ibmcloud ce application get -n dealerdetails' to check the application status.
OK
https://dealerdetails.xj562h09ws5.us-south.codeengine.appdomain.cloud
```

Please note that if you encounter the error FAILED Wait failed for application 'dealerdetails', you can rename the application to dealerdetails1 and re-execute the command.

Part B: Deploy the Frontend Microservice

1. Open new terminal, go to /home/project directory.

```
cd /home/project
```

2. Clone the repository https://github.com/ibm-developer-skills-network/dealer_evaluation_frontend.git in your /home/project directory.

Take a screenshot of the successful git cloning and save it as git_clone.png.

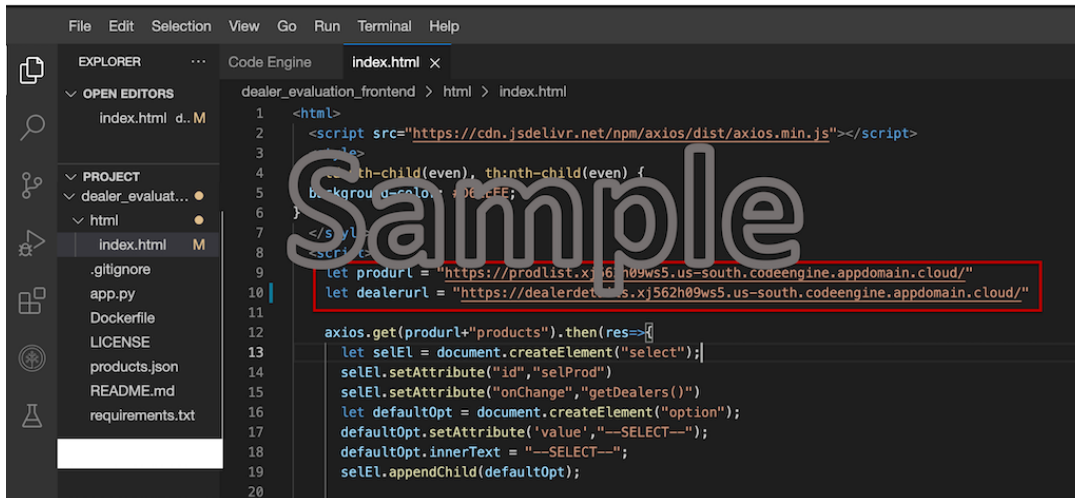


```
theia@theiadocker-lavanyas:/home/project$ git clone https://github.com/ibm-developer-skills-network/dealer_evaluation_frontend.git
Cloning into 'dealer_evaluation_frontend'...
remote: Enumerating objects: 16, done.
remote: Counting objects: 100% (16/16), done.
remote: Compressing objects: 100% (12/12), done.
remote: Total 16 (delta 2), reused 14 (delta 1), pack reused 0
Unpacking objects: 100% (16/16), done.
theia@theiadocker-lavanyas:/home/project$ cd dealer_evaluation_frontend/
```

3. Change to the dealer_evaluation_frontend directory.

4. Update the index.html file with the deployment URLs obtained from the microservice deployments. (<http://localhost:5000/> and <http://localhost:8080/>), copy the deployment URLs you copied in the appropriate location. Make sure you end the URLs with a /.

Take a screenshot of the changes and save it as index_urlchanges.png.



```
1 <html>
2 <script src="https://cdn.jsdelivr.net/npm/axios/dist/axios.min.js"></script>
3
4 <div class="container">
5   <div class="row">
6     <div class="col">
7       <div class="card">
8         <div class="card-body">
9           <div class="text">
10            <div class="text">
11              <div class="text">
12                <div class="text">
13                  <div class="text">
14                    <div class="text">
15                      <div class="text">
16                        <div class="text">
17                          <div class="text">
18                            <div class="text">
19                              <div class="text">
20                                <div class="text">
```

5. Deploy the Dealer Evaluation frontend microservice by pointing the build-source to the current directory.

build-source - .

port - 5001

name - frontend

image - us.icr.io/\${SN_ICR_NAMESPACE}/frontend

Take a screenshot of the successful deployment and name it frontend_deploy.png.

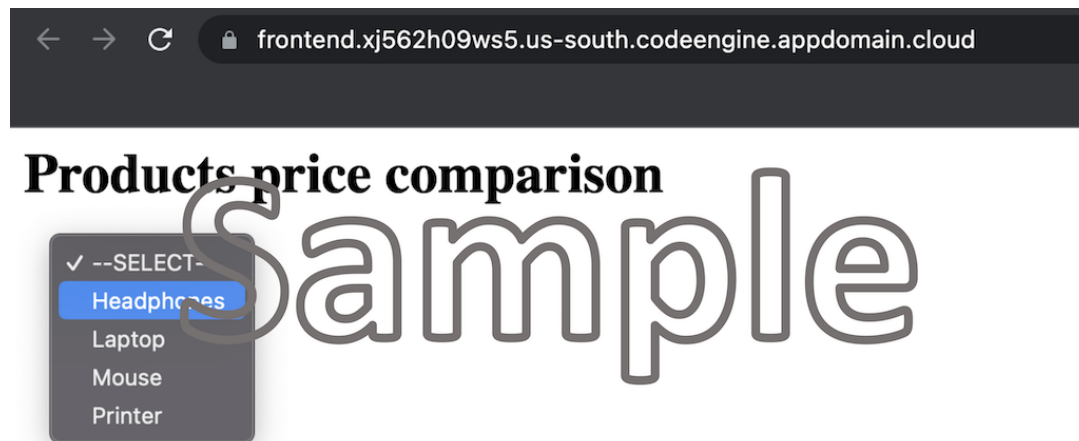
```
theia@theiadocker-lavanyas:/home/project/dealer_evaluation_frontend$ ibmcloud ce application create --name frontend --image us.icr.io/${SN_ICR_NAMESPACE}/frontend --registry-secret icr-secret --port 5001 --build-source .
Creating application 'frontend'...
Packaging files to upload from source path './'...
Submitting build run 'ibmcloud-run-230110-080653852'...
Creating image 'us.icr.io/enabs-lavanyas/frontend:230110-1306-lcccz'...
Waiting for build run to complete...
Build run status: 'running'
Build run completed successfully.
Run 'ibmcloud ce build run get -n frontend -r n-3010-0-0651852' to check the build run status.
Waiting for application 'frontend' to become ready.
Configuration 'frontend' is waiting for a Revision to become ready.
Ingress has not yet been reconciled.
Waiting for load balancer to be ready.
Run 'ibmcloud ce application get -n frontend' to check the application status.
OK
https://frontend.xj562h09ws5.us-south.codeengine.appdomain.cloud
```

Please note that if you encounter the error FAILED Wait failed for application 'frontend', you can rename the application to frontend1 and re-execute the command.

6. Click the link to load the homepage. Please note the page takes time to load the first time you access it.

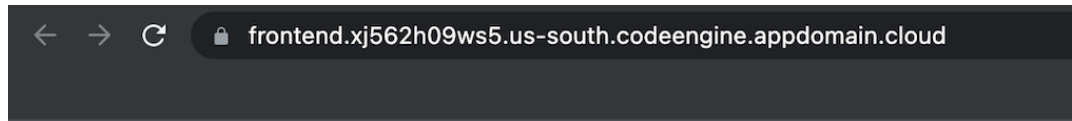
7. Click the products drop down to see if the products have been populated.

Take a screenshot of the home page showing the products list and name it homepage.png.

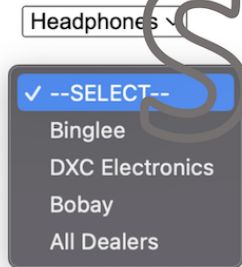


8. Choose a specific dealer for the product and verify the price is displayed.

Take a screenshot of the entire page showing the product chosen, and dealers that supply the listed product returned by the microservice and name it product_dealer.png.



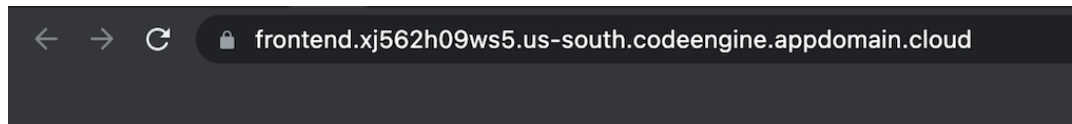
Products price comparison



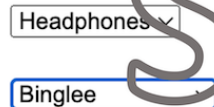
9. After the dealers dropdown populates, choose a particular dealer for the product and see if the price charged by that dealer is displayed.

Allow 10 to 20 secs to load the page.

Take a screenshot of the entire page showing the product chosen, dealer chosen, and the price returned by the microservice and name it `product_dealer_price.png`.



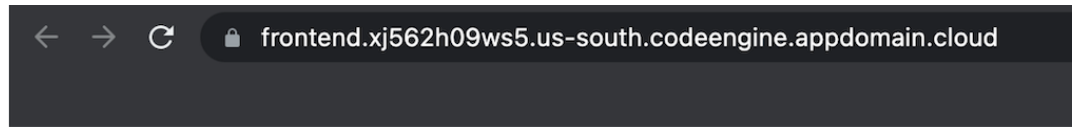
Products price comparison



Headphones costs \$30 at Binglee

10. Choose All Dealers option for a product (make sure you choose a product which has more than one dealer). Pricing of all dealers offering the product should be shown on the screen.

Take a screenshot of the entire page showing the product chosen, All Dealers option chosen, and the prices charged by all dealers returned by the microservice and name it `product_all_dealers_prices.png`.



Products price comparison

Headphones ✓

All Dealers ✓

Sample

Binglee	\$30
DXC Electronics	\$20
Bobay	\$20

Congratulations! You have completed the Final project!

Summary:

In this lab, you've successfully deployed multiple microservices to build an integrated application. This includes two backend microservices, one developed in Python and the other in Node.js, along with a front-end microservice.

Author(s)

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